

Exploring Embodied Interactive Techniques

with TouchDesigner and MediaPipe

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ABSTRACT

Creating embodied interactive experiences has historically required a host of custom hardware sensors. With developments in computer vision and accessible software solutions it is now possible to create a variety of interactive experiences using only a webcam and laptop. Our hands-on lab, "Exploring Embodied Interactive Techniques: with TouchDesigner and MediaPipe," delves into these possibilities. Throughout the lab, participants will be introduced to TouchDesigner and MediaPipe, alongside other computer vision and AI techniques. Techniques covered include, Hand / Gesture Tracking, Facial Landmark Detection, Pose tracking, and Object Tracking. This mix of techniques highlights the diverse set of inputs and outputs available to creators. Through demonstrations and activities, we will showcase the potential applications of these tools in creating interactive experiences. Participants will gain practical skills and knowledge with these freely available tools. Furthermore, examples and case studies from the presenters will add real world context to the presentation.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI).

KEYWORDS

Real-Time, Tracking, TouchDesigner, MediaPipe

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1 LAB OVERVIEW

In this hands-on lab that is open to all skill levels we will discuss and explore the use of TouchDesigner and MediaPipe to create interactive experiences that engage the human body. MediaPipe is an open source, cross-platform, customizable Machine Learning solution for live and streaming media [Lugaresi, 2019]. Combining MediaPipe with TouchDesigner, "a node-based visual programming software for real-time interactive multimedia content", we can easily use computer vision to create embodied interactions [Wikipedia,

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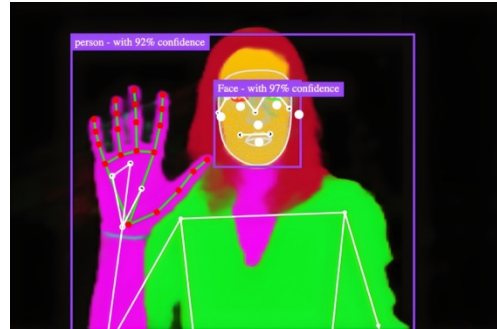


Figure 1: Example output from MediaPipe. Displaying Hand Tracking, Face Tracking, Pose Tracking, Image Segmentation and Object Detection.

2024]. With only a laptop (MacOS or Windows) equipped with a webcam we can track hands, body poses, facial expressions, and perform object tracking - all without writing any code. In this hands-on lab we will highlight different use cases for this technology and dive into setting up this system on participants' own machines.

During the lab we will build a simple painting application inside of TouchDesigner, first using mouse input and then extending the application to be used with MediaPipes' various computer vision algorithms. The drawing application will include UI interactions to configure paint color, brush width, erasing, and toggle visual effects. We will learn how to replace the mouse input with various MediaPipe inputs such as hand tracking and gaze direction while also simulating mouse clicks with gestures, movements, and hovers. Through the building of this application participants will acquire the skills to dream up and create their own interactive projects that can engage the human body. Time will be built into the end of the course for participants to ask questions and seek advice from the instructor on how they might implement these techniques into their own projects.

1.1 MediaPipe TouchDesigner Plugin

The MediaPipe TouchDesigner Plugin was developed by Torin Blankensmith and is "a GPU Accelerated, self-contained, MediaPipe Plugin for TouchDesigner that runs on Mac and PC with no installation. This project currently supports all MediaPipe vision models except Interactive Segmentation and Image Embedding" [Blankensmith, 2023]. This plugin enables TouchDesigner users' easy access to all the data from the supported MediaPipe models. With this data available we can rapidly experiment and prototype with a range of interactions driven by the human body.

1.2 Learning Outcomes

Participants will complete the lab with these takeaways.

- Understand the fundamentals of TouchDesigner and MediaPipe as tools for creating interactive experiences that engage the human body.
- Set-up and configure TouchDesigner and MediaPipe on personal laptops equipped with webcams.
- Identify and articulate different use cases for integrating TouchDesigner and MediaPipe in interactive applications.
- Apply interactive techniques using TouchDesigner and MediaPipe to track hands, body poses, facial expressions, and perform object tracking without the need for coding.

- Analyze and evaluate the potential impact of embodied interactions in various contexts such as art installations, performance art, gaming, and educational experiences.

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